CONTINUITY

Intermediate Value Theorem

The theorem only goes one way.

THE THEOREM SAYS

If f is continuous on $[a, b]$ and k lies between $f(a)$ and $f(b)$, then $f(c) = k$ for some $c \in (a, b)$.

IT DOES NOT SAY

~~If k does not lie between $f(a)$ and $f(b)$, then f never reaches k on $[a, b]$.~~

READING THE THEOREM BACKWARDS

WRONG WORKING

$f(x) = x^2$ on $[-1, 1]$ has $f(-1) = f(1) = 1$. There is no value 0 between them, so f never takes the value 0.

RIGHT WORKING

The theorem speaks only about values between $f(a)$ and $f(b)$. Here $f(0) = 0$.

Existence only: it says a c exists but does not tell you where or whether it is unique.